

MAT 523: Functions of a Real Variable I

Homework II

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(1.)

Theorem

Let a and b be real numbers.

- (i) If $ab = 0$, then $a = 0$ or $b = 0$.
- (ii) $a^2 - b^2 = (a - b)(a + b)$ and if $a^2 = b^2$, then $a = b$ or $a = -b$.
- (iii) Let c be a positive real number. Define $E = \{x \in \mathbb{R} | x^2 < c\}$. Verify that E is nonempty and bounded above. Define $z = \sup E$. Show that $z^2 = c$. Use part (ii) to show that there is a unique $x > 0$ for which $x^2 = c$. It is denoted by $\sqrt{c}$.

Solution

- (i) Suppose $a \neq 0$ and $b \neq 0$.

Case 1: Suppose $a > 0$ and $b > 0$. Then, by the first positivity axiom, $ab > 0$, hence $ab \neq 0$.

Case 2: Suppose $a < 0$ and $b < 0$. Then $(-a)(-b) > 0$. But

$$\begin{aligned}(-a)(-b) &= [(-1)a][(-1)b] \\&= (-1)^2(ab) \\&= 1ab \\&= ab.\end{aligned}$$

Therefore, $ab > 0$, so $ab \neq 0$.

Case 3: Suppose, without loss of generality, that $a > 0$ and $b < 0$. Then $a(-b) > 0$. But

$$\begin{aligned}a(-b) &= a[(-1)b] \\&= (-1)[ab] \\&= -ab.\end{aligned}$$

So $-ab > 0$, which implies that $ab < 0$, i.e., $ab \neq 0$.

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(ii)

$$\begin{aligned}a^2 - b^2 &= [a^2 + 0] - b^2 \\&= [a^2 + (ab - ab)] - b^2 \\&= [a^2 + ab] + [-(ab) - b^2] \\&= a(a + b) + [(-1)ab + (-1)b^2] \\&= a(a + b) + (-1)[ab + b^2] \\&= a(a + b) - b(a + b) \\&= (a + b)a + (a + b)(-b) \\&= (a + b)(a - b).\end{aligned}$$

(iii) Note that $0^2 = 0$ is less than any positive real number c , so $0 \in E$. Thus $E \neq \emptyset$. To show that E is bounded above, we consider two cases.

Case 1: Suppose $c \in (0, 1)$. Then $\frac{1}{c} > 1$. Note that if $x^2 < c$, then $-1 < x < 1$. Thus, E is bounded above by $\frac{1}{c}$.

Case 2: Suppose $c \in [1, \infty)$. If $x^2 < c$, then it is also true that $-c < x < c$. Thus, E is bounded above by c .

To show that $z^2 = c$, we first show that E contains positive elements.

(2.)

Theorem

The irrational numbers are dense in $\mathbb{R}$.

Proof

Let a, b be real numbers with $b > a$. Then $\frac{b}{\sqrt{2}} > \frac{a}{\sqrt{2}}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists a rational number r satisfying

$$\frac{b}{\sqrt{2}} > r > \frac{a}{\sqrt{2}}.$$

Multiplying the above inequality by $\sqrt{2}$ yields

$$b > r\sqrt{2} > a,$$

since $\sqrt{2} > 0$. We now show that $r\sqrt{2}$ is irrational. If not, then we can write

$$r\sqrt{2} = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$ and $q \neq 0$. Furthermore, since r is rational, we can write $r = \frac{s}{t}$, where $s, t \in \mathbb{Z}$, and $t \neq 0$.

Thus,

$$\frac{s}{t}\sqrt{2} = \frac{p}{q},$$

i.e.,

$$\sqrt{2} = \frac{pt}{qs}.$$

But $\frac{pt}{qs}$ is rational, which contradicts the fact that $\sqrt{2}$ is irrational. Therefore, $r\sqrt{2}$ is irrational. Hence, the irrational numbers are dense in $\mathbb{R}$. ■

(5.)

Theorem

A nondegenerate interval of real numbers fails to be finite.

Proof

Let I be a nondegenerate interval of real numbers. Suppose I is finite. Since I is nondegenerate, it is nonempty. Therefore, there must be some $n \in \mathbb{N}$ such that I is equipotent to the set $\{1, 2, 3, \dots, n\}$, which means we can list every element of I (in ascending order):

$$i_1, i_2, i_3, \dots, i_n.$$

Now, take i_1 and i_2 . Since the elements of I were listed in ascending order, we have $i_2 > i_1$. By virtue of $\mathbb{Q}$ being dense in $\mathbb{R}$, it must be the case that there is an $r \in \mathbb{Q}$ satisfying $i_1 < r < i_2$. Clearly, r was not in our enumeration of I above. But by the definition of an interval of real numbers, r must be in I . This is a contradiction. Therefore, I is not finite.

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(6.)

Theorem

Suppose X is infinite. Then X is equipotent with $X \cup \{\alpha\}$.

Proof

If X is countable, then its elements can be listed:

$$x_1, x_2, x_3, \dots$$

The elements of $X \cup \{\alpha\}$ can also be listed:

$$\alpha, x_1, x_2, x_3, \dots$$

Define $f : X \longrightarrow X \cup \{\alpha\}$ by the function

$$f(x_n) = \begin{cases} \alpha & \text{if } n = 1 \\ x_{n-1} & \text{if } n \in \mathbb{N} \setminus \{1\}. \end{cases}$$

Clearly, f is a bijection. Therefore, X is equipotent with $X \cup \{\alpha\}$.

If X is uncountable, we can construct a countable subset of X ,

$$C = \{x_1, x_2, x_3, \dots\} \subset X.$$

Let $f : X \longrightarrow X \cup \{\alpha\}$ be defined as follows: **(1)** On C ,

$$f(x_n) = \begin{cases} \alpha & \text{if } n = 1 \\ x_{n-1} & \text{if } n \in \mathbb{N} \setminus \{1\}. \end{cases}$$

(2) On $X \setminus C$,

$$f(x) = x.$$

Then f is a bijection. Therefore, X is equipotent with $X \cup \{\alpha\}$.

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(7.)

Theorem

Any two nondegenerate intervals of real numbers are equipotent.

Proof

Let a, b, c, d be real numbers such that $a < b$ and $c < d$. Consider the open intervals (a, b) and (c, d) . Both are infinite, by Problem 5. Applying the result of Problem 6 to both intervals, we see that (a, b) is equipotent to $[a, b)$, $(a, b]$, and $[a, b]$. Since equipotence is an equivalence relation, all of the above intervals are equipotent to each other. Similarly, (c, d) is equipotent to $[c, d)$, $(c, d]$, and $[c, d]$, and these intervals are all equipotent to each other (again, by virtue of equipotence being an equivalence relation). If we can show that $[a, b]$ is equipotent to $[c, d]$, we will be able to conclude that all the intervals listed above are equipotent to each other. To this end, we construct a bijection between $[a, b]$ and $[c, d]$. Consider the function $f : [a, b] \rightarrow [c, d]$ defined by

$$f(x) = \left(\frac{c-d}{a-b} \right) x + \frac{ad-bc}{a-b}.$$

Clearly, f is one-to-one. To show that f is onto, let $y \in [c, d]$. Solving for x in the equation

$$\left(\frac{c-d}{a-b} \right) x + \frac{ad-bc}{a-b} = y$$

yields (after many tedious manipulations)

$$x = \frac{b(y-c) - a(y-d)}{d-c}.$$

We need to show that this is an element of $[a, b]$. If not, then either

$$\frac{b(y-c) - a(y-d)}{d-c} < a$$

or

$$\frac{b(y-c) - a(y-d)}{d-c} > b.$$

But from the first inequality, we can conclude that $y < c$, which is impossible since $y \in [c, d]$ and must therefore satisfy $y \geq c$. From the second inequality, we can conclude that $y > d$, which is also impossible, since $y \leq d$. Thus,

$$x = \frac{b(y-c) - a(y-d)}{d-c} \in [a, b].$$

Therefore, f is onto and, hence, a bijection. This means $[a, b]$ is equipotent to $[c, d]$. Thus, all of the intervals listed above are equipotent to each other. ■